

Teorema de Euler-Lagrange

Teorema 1 (Euler-Lagrange). Sea $f \in \mathcal{C}^2([a, b] \times \mathbb{R} \times \mathbb{R})$ e $y \in \mathcal{C}^2([a, b])$ mínimo o máximo de la integral $J[y] = \int_a^b f(x, y(x), y'(x)) dx$, con $y(a) = y_a$ y $y(b) = y_b$

$$\implies \forall x \in [a, b] : f_y(x, y(x), y'(x)) - \frac{d}{dx} (f_{y'}(x, y(x), y'(x))) = 0.$$

Demostración: Tomamos $\varphi \in \mathcal{C}^2([a, b])$ con $\varphi(a) = 0$ y $\varphi(b) = 0$ y consideramos

$$J[y + \varepsilon\varphi] = \int_a^b f(x, y(x) + \varepsilon\varphi(x), y'(x) + \varepsilon\varphi'(x)) dx =: \bar{J}(\varepsilon), \text{ para } \varepsilon \in \mathbb{R}.$$

La función $\bar{J}: \varepsilon \mapsto \bar{J}(\varepsilon)$ tiene un máximo o mínimo en $\varepsilon = 0$ por hipótesis, por tanto su derivada se anula en $\varepsilon = 0$:

$$\begin{aligned} 0 &= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_a^b f(x, y(x) + \varepsilon\varphi(x), y'(x) + \varepsilon\varphi'(x)) dx \\ &= \int_a^b \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} f(x, y + \varepsilon\varphi, y' + \varepsilon\varphi') dx = \int_a^b (f_y\varphi + f_{y'}\varphi') dx \end{aligned}$$

donde $f_y = \frac{\partial f}{\partial y}$ y $f_{y'} = \frac{\partial f}{\partial y'}$. Ahora bien, como $\int_a^b f_{y'}\varphi' dx = f_{y'}\varphi|_a^b - \int_a^b \frac{d}{dx} (f_{y'}) \varphi dx$

$$\implies \int_a^b f_y\varphi dx + \cancel{f_{y'}\varphi|_a^b}^0 = \int_a^b \frac{d}{dx} (f_{y'}) \varphi dx \implies \int_a^b \left(f_y - \frac{d}{dx} (f_{y'}) \right) \varphi dx = 0.$$

Por tanto, por el ~~lem~~-dubois-reymond/Lema 1, $\forall x \in [a, b] : f_y - \frac{d}{dx} (f_{y'}) = 0$. ■

Observación 1. Como $\frac{d}{dx} (f_{y'}) = f_{xy'} + y' f_{yy'} + y'' f_{y'y'}$, si $f_{y'y'} \neq 0$, podemos despejar y'' :

$$y'' = \frac{f_y - f_{xy'} - y' f_{yy'}}{f_{y'y'}}.$$

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